

Vessel Autonomy Interface Control Document — M&A / D&S Extract

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NOTE: This document is extracted from VA-ICD-0001 Rev. 0.2, containing only the Main & Auxiliary System (M&A) and Damage & Safety System (D&S) interface definitions. For the full ICD including VMS, Navigation Autonomy, and Mission Autonomy interfaces, refer to VA-ICD-0001.

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Role	Name	Signature / Date
Prepared By		
Writer	Seonghee Cho, Hyewon Chun	
Editor	Sangmun Lee	
Review & Approval		
Reviewer	Hansol Kim, Sangjeo Yeo	
Approver	Jeeyeon Jeon	

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1. Revision History

Version	Date	Description
0.1 (Draft)	2026-03-23	Initial draft (full ICD)
0.2	2026-03-26	Added M&A/D&S API schemas, JSON examples, health check
0.2-MnA-DnS	2026-03-30	Extracted M&A/D&S content into standalone document

2. Overview

2.1 Purpose & Scope

This document defines the interface between Vessel Autonomy (VA) and two Machinery Autonomy sub-systems targeting the **Alpha vessel**:

- **Main & Auxiliary System (M&A)**: Vessel machinery diagnostics (engines, generators, auxiliary equipment) and operational advisories
- **Damage & Safety System (D&S)**: CCTV-based vessel and machinery damage detection (fire, smoke, flooding, leakage, soot/burn marks, structural damage) and intrusion/threat monitoring

Items applicable only to the Surrogate vessel demonstration are explicitly marked as **(Surrogate only)** or **(Surrogate Demo)** throughout the document.

This document specifies:

- VA's transmit (Tx) and receive (Rx) responsibilities for M&A and D&S
- API endpoint definitions and JSON schemas
- Data flow between M&A/D&S, VA, and Mission Autonomy (MA)

2.2 Document References

Reference	Title	Notes
VA-ICD-0001	Vessel Autonomy Interface Control Document (Full)	Parent document

2.3 Terminology

Term	Definition
VA	Vessel Autonomy - the supervisory integration system
M&A	Main & Auxiliary System - vessel machinery diagnostics and operational advisories
D&S	Damage & Safety System - CCTV-based damage detection and threat monitoring
MA	Mission Autonomy - mission planning and tasking system
VMS	Vessel Management System (HiCONiS) - vessel platform control layer

3. Communication Architecture (Proposed -- TBD)

The following communication architecture is proposed but not yet finalized.

3.1 Proposed Protocol Strategy

A hybrid protocol model is proposed based on data characteristics:

Protocol	Use Case	Applicable Signals
REST API (HTTP POST)	Event-driven, critical messages requiring delivery confirmation	OperationalAdvisory, DetectEventResult, ResolveEventResult
REST API (HTTP POST)	Periodic status with structured JSON payload	DiagnosticResult, healthCheck
UDP Unicast	High-frequency, real-time, loss- tolerant data	DPS Thruster Feedback (Message 1100)

Rationale: REST API provides built-in delivery confirmation via HTTP response, which serves as implicit ACK for critical signals such as OperationalAdvisory. UDP is reserved for high-frequency relay data (DPS Thruster Feedback) where occasional packet loss is acceptable as the next update arrives within ~1 second.

3.2 VA Receive Endpoints (Proposed)

The following REST API endpoints are proposed for VA to receive data from M&A and D&S:

M&A -> VA Endpoints:

Method	Endpoint (TBD)	Signal	Description
POST	/va/api/v1/ma/diagnostic- result	DiagnosticResult	Periodic machinery diagnostic report
POST	/va/api/v1/ma/ operational-advisory	OperationalAdvisory	Event-driven operational advisory (ACK via HTTP 200)

D&S -> VA Endpoints:

Method	Endpoint (TBD)	Signal	Description
POST	/va/api/v1/ds/detect-event	DetectEventResult	Damage detection event report
POST	/va/api/v1/ds/resolve-event	ResolveEventResult	Damage resolution event report
POST	/va/api/v1/ds/health-check	healthCheck	Camera connection status
POST	/va/api/v1/ds/threat-detection	Threat Detection	External threat detection report

UDP Data (DP -> VA -> M&A):

Protocol	Port (TBD)	Multicast/Unicast (TBD)	Signal	Description
UDP	TBD	TBD		

Protocol	Port (TBD)	Multicast/Unicast (TBD)	Signal	Description
			DPS Thruster Feedback	DP Message 1100 relay (~1 Hz)

3.3 VA Transmit Endpoints (Proposed)

The following endpoints are proposed for VA to send commands to M&A and D&S sub-systems:

Method	Endpoint (TBD)	Target	Description
POST	/ma/api/v1/diagnostic-request	M&A	Request on-demand diagnostic run
POST	/ma/api/v1/advisory-response	M&A	OperationalAdvisory accept/reject response
POST	/ds/api/v1/command (TBD)	D&S	D&S downlink commands (TBD)

NOTE: The above communication architecture is a proposal for team discussion. Key items requiring agreement: - Protocol confirmation: REST API vs. MQTT vs. gRPC for production - Port assignments for all endpoints - Authentication/authorization mechanism - API versioning strategy - Error handling and retry policy - Network topology: direct connection vs. message broker

4. Main & Auxiliary System (M&A)

The Main & Auxiliary System provides engine/equipment diagnostics and operational advisory capabilities.

4.1 Downlink (VA -> M&A)

VA sends operational advisory responses and relays DP thruster feedback to the M&A system.

NOTE: This interface is specific to the Surrogate vessel demonstration only and is not part of the standard VA-M&A interface. DP Thruster Feedback (Message 1100) provides relative thruster parameters only -- it does not include engine RPM, load, or torque data. Transport: UDP (port and rate details TBD).

#	Signal	Description	Trigger
1	OperationalAdvisory Response	Response to operational advisory with accept/reject	On advisory received

#	Signal	Description	Trigger
2	DPS Thruster Feedback	Thruster feedback (Message 1100) relayed from DP system	UDP (relay)

Device ID Mapping:

DeviceID	Equipment
1 - 2	Bow Thruster
3 - 6	Waterjet

Data Flow: DP System -> VA (UDP) -> M&A

Thruster feedback data received from the Dynamic Positioning (DP) system via UDP. VA relays DP Message 1100 to M&A for propulsion monitoring. This is the only real-time propulsion-related data available from the vessel's DP system -- it does not include engine RPM, load, or torque.

DPS Thruster Feedback Fields:

Field	Type	Range	Description
DeviceID	Integer	1 ~ 12	Thruster device identifier (1-2: Bow Thruster, 3-6: Waterjet)
RPM_Fba	Float	0.0 ~ 1.0	Relative RPM feedback (0~100%)
Pitch_Fba	Float	0.0 ~ 1.0	Relative pitch feedback (waterjet: reversing bucket)
Angle_Fba	Float	0 ~ 360	Thruster azimuth angle (degrees)

4.1.1 OperationalAdvisoryResponse

Response to an operational advisory received from M&A. VA sends accept (0) or reject (1) for each advisory.

JSON Schema Example:

```
{
  "advisoryId": 1001,
  "response": 0,
  "timestamp": "2026-01-14T10:05:00Z"
}
```

Response values: - **0** -- **Accept:** VA accepts the operational advisory recommendation - **1** -- **Reject:** VA rejects the operational advisory recommendation

4.1.2 DPS Thruster Feedback (Relay)

NOTE: DPS Thruster Feedback is specific to the Surrogate vessel demonstration only.

4.2 Uplink (M&A -> VA)

4.2.1 DiagnosticResult

Periodic machinery diagnostic results per equipment. Sent at configurable intervals.

JSON Schema Example:

```
{
  "save_timestamp": "2026-01-14T10:30:00Z",
  "save_interval_seconds": 10,
  "equipment_count": 4,
  "operation_mode": "GT",
  "results": {
    "ME1": {
      "PRIMARY": {
        "timestamp": "2026-01-14T10:29:30Z",
        "status": "RUNNING",
        "result": "NORMAL",
        "score": 95,
        "updated_by": "model_v2"
      },
      "SECONDARY": {
        "Intake": { "result": "NORMAL", "score": 98, "detail": {} },
        "Exhaust": { "result": "NORMAL", "score": 92, "detail": {} }
      }
    },
    "ME2": {...}, "ME3": {...}, "ME4": {...}
  }
}
```

DiagnosticResult Fields:

#	Field	Type	Description
1	save_timestamp	String (ISO 8601)	Diagnostic execution timestamp
2	save_interval_seconds	Integer	Diagnostic interval in seconds
3	equipment_count	Integer	Number of diagnosed equipment
4	operation_mode	Enum (GT/EPM/ null)	Gas turbine operation mode (null if not applicable)

#	Field	Type	Description
5	results	Dictionary	Per-equipment diagnostic results (ME1, ME2, ME3, ME4)

Each equipment result contains two diagnosis levels:

Level	Fields	Description
PRIMARY	timestamp, status, result, score, updated_by	Primary diagnosis - overall equipment health score
SECONDARY	timestamp, status, result, score, updated_by, subunit_score	Secondary diagnosis - per-subunit breakdown with detail

SECONDARY subunit categories (per ME):

- Intake, Exhaust, Turbine, Cooling, Fuel, Lubrication, Control

Each subunit_score contains:

Field	Type	Description
result	Enum	NORMAL / CAUTION / WARNING / DANGER / STOP
score	Float	Diagnostic score (0~100)
detail	Dictionary	Detailed anomaly info: tag, deviation, duration, etc.

Diagnostic result levels:

Result	Description
NORMAL	Equipment operating within normal parameters
CAUTION	Minor anomaly detected, monitoring recommended
WARNING	Significant anomaly, corrective action may be needed
DANGER	Critical condition, immediate action required
STOP	Equipment stopped or non-operational

NOTE: Surrogate vessel: 4 main engines (ME1~ME4), 9 visualization equipment (Waterjet x4, GE x3, Bowthruster x2).

4.2.2 Operational Advisory

Operational advisory generated when machinery issues require vessel-level action. Contains recommended action options.

JSON Schema Example:

```
{
  "metadata": {
    "request_id": "8a1a5d44-1a56-4c7f-9b6b-2f7b0b8a5f90",
    "timestamp": "2026-03-04T13:20:00Z",
    "ship_id": "IMO-9876543",
    "version": "v1.0"
  },
  "machinery_status": {
    "overall_status": "WARNING",
    "detected_issues": [
      {
        "machine_name": "ME1",
        "severity": "WARNING",
        "alarms": [
          {"message": "High exhaust temperature", "duration_sec": 120}
        ]
      }
    ]
  },
  "operational_advisory": {
    "reason_summary": "ME1 exhaust temperature exceeding threshold",
    "options": [
      {
        "type": "SHIP_SPEED_REDUCTION",
        "is_recommended": true,
        "cause_machines": ["ME1"],
        "description": "Reduce speed by 30%"
      },
      {
        "type": "REDUCED_POWER_RTb",
        "is_recommended": false,
        "cause_machines": [],
        "description": "Return to base"
      }
    ]
  }
}
```

OperationalAdvisory Fields:

#	Section	Field	Type	Description
1	metadata	request_id	String (UUID v4)	Unique request identifier
2	metadata	timestamp	String (ISO 8601)	Advisory generation timestamp

#	Section	Field	Type	Description
3	metadata	ship_id	String	Ship identifier (e.g., IMO number)
4	metadata	version	String	API version
5	machinery_status	overall_status	Enum	NORMAL / CAUTION / WARNING / DANGER
6	machinery_status	detected_issues	Array	List of detected equipment issues
7	operational_advisory	reason_summary	String	Summary of situation and cause
8	operational_advisory	options	Array	Recommended action options

detected_issues (per equipment):

Field	Type	Description
machine_name	String	Equipment identifier (e.g., ME1, GE2)
severity	Enum	CAUTION / WARNING / DANGER
alarms	Array	Active alarms: message (String), duration_sec (Integer)

operational_advisory.options:

Each option contains:

Option Type	Description
SHIP_SPEED_REDUCTION	Reduce vessel speed (expressed as reduction ratio)
SHIP_STOP	Emergency stop
REDUCED_POWER_RTb	Return to base with reduced power

Field	Type	Description
type	Enum	SHIP_SPEED_REDUCTION / SHIP_STOP / REDUCED_POWER_RTb
is_recommended	Boolean	Whether this option is the recommended action
cause_machines	Array	Equipment IDs causing this advisory (e.g., ["ME1", "ME2"])
description	String	Detailed description of the action

NOTE (Surrogate Demo Only): The current M&A Operational Advisory definition allows specifying an absolute target speed in the description field for SHIP_SPEED_REDUCTION (e.g., "Reduce speed to 10 kts"). However, in the Surrogate vessel demonstration environment, retrieving real-time vessel speed from the Surrogate system into M&A is limited,

making it infeasible to derive a concrete target speed for reduction. Therefore, for the Surrogate Demo, one of the following approaches shall be applied:

1. **Remove absolute target speed** from the SHIP_SPEED_REDUCTION description, or
2. **Express reduction as a relative ratio** (e.g., "Reduce speed by 50%") instead of an absolute target speed.

This constraint applies **only to the Surrogate Demo** and does not affect the general ICD specification for production vessels where real-time speed data may be available.

5. Damage & Safety System (D&S)

The Damage & Safety System analyzes onboard CCTV video streams to detect damage events and security threats.

D&S provides two detection capabilities:

- **Damage Detection:** Internal camera analysis for fire, smoke, flooding, leakage, soot/burn marks, structural damage
- **Threat Detection:** External camera analysis for intruder detection (unauthorized personnel, deck monitoring, hatch-lock status)

5.1 Downlink (VA -> D&S)

VA relays sensor data received from VMS to the D&S system. Protocol: Modbus/TCP (TBD).

#	Signal	Description	Trigger
1	Smoke Detector Sensor Data	Smoke detector sensor data relayed from VMS (VMS -> VA -> D&S)	Periodic (relay)
2	Fire Detector Sensor Data	Fire detector sensor data relayed from VMS (VMS -> VA -> D&S)	Periodic (relay)

NOTE: Sensor data originates from VMS and is relayed through VA to D&S (VMS -> VA -> D&S). Communication protocol is planned as Modbus/TCP (TBD).

5.2 Uplink (D&S -> VA)

5.2.1 DetectEventResult (Damage Detection)

Sent when a disaster or safety event is detected by internal camera-based analysis.

JSON Schema Example:

```
{
  "cameraId": 1,
  "eventId": 3,
  "eventClass": "Disaster/Accident",
  "eventName": "Fire",
  "eventLevel": 4,
  "savePath": "1/fire/imagepath.png",
  "createdAt": "2026-01-14T10:00:00Z",
  "contamination": 1,
  "detectionInfo": {
    "classId": 0,
    "className": "fire",
    "weight": "136.pth",
    "xyxy": [0.3640625, 0.2680555, 0.421875, 0.3277777]
  },
  "anomalyMap": "1/fire/anomalyMapimagepath.npy",
  "anomalyScore": 0.94
}
```

Detectable damage types:

Type	Description
Fire	Active fire detected
Smoke	Smoke presence detected
Flooding	Water ingress detected
Leakage	Fluid leakage detected
Soot / Burn Marks	Post-fire residue or burn damage
Structural Damage	Visible structural damage to vessel components

NOTE: In the Surrogate vessel demonstration, only Fire and Smoke events are supported for detection. Other damage types (Flooding, Leakage, Soot/Burn Marks, Structural Damage) will be integrated in subsequent phases.

DetectEventResult Fields:

#	Field	Type	Description
1	cameraId	Integer	Camera identifier
2	eventId	Integer	Unique event identifier. Used by ResolveEventResult to reference this detection event.
3	eventClass	String	Event classification (e.g., "Disaster/Accident")

#	Field	Type	Description
4	eventName	String	Event name (e.g., "Fire")
5	eventLevel	Integer (1~4)	Risk level (1=lowest, 4=highest)
6	savePath	String	Event image save path
7	createdAt	Timestamp (ISO 8601)	Event detection timestamp
8	contamination	Integer (0/1)	Camera contamination flag
9	detectionInfo	Dictionary	Detected object details (classId, className, weight, xyxy)
10	anomalyMap	String	Highlighted abnormal-area heatmap image path
11	anomalyScore	Float	Anomaly confidence score (0.0~1.0)

detectionInfo fields:

Field	Type	Description
classId	Integer	Object class ID
className	String	Object class name (e.g., "fire")
weight	String	Detection model name
xyxy	Dictionary	Bounding box coordinates [x1, y1, x2, y2] (normalized)

Additional information provided with damage events:

- Location of occurrence and time of occurrence
- Time of action completion
- Detected object information

5.2.2 Threat Detection (Tentative)

The system analyzes video streams from externally mounted CCTV cameras to detect potential security threats.

Category	Detection Method	Primary Targets	Information Provided
Intruder Detection	External surveillance CCTV (incl. night-vision), area monitoring	Unauthorized personnel entering deck area	Initial detection time, final disappearance time, proximity to hatch

NOTE: Threat Detection is tentative and subject to further testing and integration validation. This capability is not included in the Surrogate vessel demonstration scope and will be evaluated in subsequent integration phases.

5.2.3 ResolveEventResult

Sent when a previously detected damage event is resolved (ended).

JSON Schema Example:

```
{
  "cameraId": 1,
  "eventId": 3,
  "endedAt": "2026-01-14T10:03:00Z",
  "contamination": 1,
  "detectionInfo": {
    "classId": null,
    "className": null,
    "weight": null,
    "xyxy": null
  },
  "anomalyMap": "1/fire/anomalyMapimagepath.npy",
  "anomalyScore": 0.12
}
```

ResolveEventResult Fields:

#	Field	Type	Description
1	cameraId	Integer	Camera identifier
2	eventId	Integer	Reference to original DetectEventResult
3	endedAt	Timestamp (ISO 8601)	Event end timestamp
4	contamination	Integer (0/1)	Camera contamination flag
5	detectionInfo	Dictionary	Detection info at resolution (fields may be NULL)
6	anomalyMap	String	Anomaly heatmap image path at resolution
7	anomalyScore	Float	Anomaly score at resolution

NOTE: When an event resolves, detectionInfo fields (classId, className, xyxy) may be NULL, indicating the detected object is no longer present.

5.2.4 healthCheck (Camera Connection Monitoring)

Periodic health check signal from the D&S system indicating connection status of all cameras. Used for monitoring camera availability, position, and detecting disconnections across the entire camera system.

JSON Schema Example:

```
{
  "cameras": [
    {
      "cameraId": 1,
      "cameraName": "Engine Room Cam 1",
      "cameraStatus": "Connected",
      "position": "Engine Room"
    },
    {
      "cameraId": 2,
      "cameraName": "Bridge CCTV",
      "cameraStatus": "Disconnected",
      "position": "Bridge"
    }
  ],
  "timestamp": "2026-01-14T10:00:00Z"
}
```

healthCheck Fields:

Field	Type	Description
cameraId	Integer	Camera identifier
cameraName	String	Camera display name
cameraStatus	String	Connection status ("Connected" / "Disconnected")
position	String	Camera installation location
timestamp	Timestamp (ISO 8601)	Health check timestamp

NOTE: healthCheck enables VA to track camera availability across D&S. Disconnection events should be relayed to MA for operator awareness.

6. M&A / D&S Signal Relay to Mission Autonomy

VA acts as a relay hub between M&A/D&S and Mission Autonomy (MA). This section describes how M&A/D&S signals are forwarded to/from MA through VA.

6.1 Downlink (MA -> VA -> M&A/D&S)

Commands from MA relayed through VA to M&A/D&S sub-systems.

#	Signal	Description	Downstream Target	Trigger
1	OperationalAdvisory Response	Accept/reject response to operational advisory from M&A	M&A API (JSON)	On advisory received
2	Emergency Stop Command	Emergency stop command (TBD)	TBD	On operator command

NOTE: OperationalAdvisory Response is relayed from MA operator decision to M&A via VA. Emergency stop command format and protocol are TBD pending MA <-> VA protocol finalization.

6.2 Uplink (M&A/D&S -> VA -> MA)

The following M&A/D&S status signals are relayed to MA for operator monitoring. All signals originate from M&A/D&S sub-systems and are received by VA via the interfaces defined in Chapters 4 and 5.

M&A Signals:

#	Signal	Description	Upstream Source	Period
1	DiagnosticResult	Periodic machinery diagnostic results per equipment	M&A API (JSON)	Configurable (default 10s)
2	OperationalAdvisory	Operational advisory with recommended actions when machinery issue detected	M&A API (JSON)	Event-driven

D&S Signals:

#	Signal	Description	Upstream Source	Period
3	DetectEventResult	Disaster/safety event detection from onboard CCTV	D&S API (JSON)	Event-driven

#	Signal	Description	Upstream Source	Period
4	ResolveEventResult	Resolution notification for previously detected event	D&S API (JSON)	Event-driven
5	healthCheck	Camera connection status monitoring	D&S API (JSON)	Periodic
6	Threat Detection	External threat detection from CCTV cameras	D&S API (JSON)	Event-driven

Proposed MA Relay Channel Allocation:

Channel	Signal	Delivery
Channel A (Periodic/ UdPbC)	DiagnosticResult, healthCheck	No ACK required
Channel B (Event-driven/RaUDP)	OperationalAdvisory, DetectEventResult, ResolveEventResult, Threat Detection	ACK required

7. Signal Summary

7.1 Main & Auxiliary System

Direction	Signals	Count
Downlink (VA -> M&A)	OperationalAdvisoryResponse, DPS Thruster Feedback	2
Uplink (M&A -> VA)	DiagnosticResult, OperationalAdvisory	2
Subtotal		4

7.2 Damage & Safety System

Direction	Signals	Count
Downlink (VA -> D&S)	Smoke Detector Sensor Data, Fire Detector Sensor Data	2
Uplink (D&S -> VA)	DetectEventResult, ResolveEventResult, Threat Detection, healthCheck	4
Subtotal		6

7.3 Grand Total (M&A + D&S)

Sub-system	Downlink	Uplink	Total
Main & Auxiliary System	2	2	4
Damage & Safety System	2	4	6
Grand Total	4	6	10

8. Open Items & TBD

#	Item	Description
1	M&A downlink format	Diagnostic request and advisory acknowledgement message format
2	D&S downlink format	Camera configuration, event acknowledgement, and detection mode control format
3	Diagnostic interval	Default diagnostic interval and on-demand trigger mechanism
4	Event level mapping	Mapping of D&S eventLevel (1~4) to VA alert severity levels
5	Advisory action flow	How VA processes OperationalAdvisory and forwards commands to equipment
6	Threat Detection	Threat Detection API format and integration scope
7	Transport protocol	Protocol confirmation: REST API vs. MQTT vs. gRPC for production
8	Port assignments	Port assignments for all M&A/D&S endpoints
9	Authentication	Authentication/authorization mechanism for REST API
10	OperationalAdvisory speed expression	Surrogate Demo에서 현재 선속 정보를 M&A가 받아올 수 없어 절대 목표 선속 명시 불가 — 삭제 또는 감소 비율(e.g., 50% 감소)로 대체 필요